

Walkthrough test script

Every screen, on the machines the demonstration uses, ticked by a person. The automated walks already proved each claim on this build; this is your own hands and eyes. About ninety minutes the first time, forty after.

Tick a line only when you saw it. If it does not match, press *Didn't match* and write what you saw — that is the bug report. Ticks stay in this browser only.

Laptop / projector (the Mac)
`https://localhost:3200`

iPad (same hotspot or Wi-Fi) — use the name, it never changes
`https://pratiks-macbook-air-3.local:3200`

Any other phone on the same network
`bash scripts/rehearsal.sh status`
prints the number

`inspector1` `supervisor1` `drm1` `admin1` `password: password123`

In this order: Pixel hotspot on → Mac and iPad join it → `bash scripts/rehearsal.sh start` → `status`. **Once, on the iPad:** `bash scripts/rehearsal.sh cert` puts *WRS-Raipur-workshop-CA.crt* on the Desktop; AirDrop it → Settings → Profile Downloaded → Install → then Settings → General → About → Certificate Trust Settings → switch on *WRS Raipur workshop CA*. After that no warning on any network, and the camera and offline use are allowed. The Pixel that runs the hotspot cannot open the address itself; the iPad scans the QR. Mobile data off on the iPad. Do this the evening before, never in the room.

1. Before signing in

0 / 5 [Clear this section](#)

Laptop and tablet, no account yet



1.1

Open the address on the laptop

Sign-in page; the हिंदी toggle top right; no blank page, no error

☐	1.2	Tap हिंदी, then English	Every label on the sign-in page changes language, both ways
☐	1.3	Sign in with a wrong password	A plain refusal in the chosen language; nothing else leaks
☐	1.4	First install the certificate on the tablet (TABLET_TRUST.md) and turn its mobile data off. Then open the address	Same page with no warning; the browser offers "Add to home screen" — accept it. Without the certificate the page opens but keeps nothing for offline use and the camera stays shut
☐	1.5	Tablet: Wi-Fi off, reopen the app icon	It still opens; an offline indicator shows. Wi-Fi back on → reconnects within seconds. If it "stops loading" after Wi-Fi returns, the tablet has switched to mobile data — see TABLET_TRUST.md

2. Inspector — on the tablet

0 / 22

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Sign in as `inspector1`. Every step below, with the phone screenshot and the exact taps: [Inspector walk](#).

☐	2.1	Home	Two large cards, both always visible: Springs (sort on the bench, or measure one) and A wagon (checklist, wheels, photographs, parts). Directly under them: Your record today — count, bands and the day's list, newest first
☐	2.2	Sorting	Bogie type, Used/New, Outer/Inner/Snubber; the strip: six coloured buttons for a used spring (Blue, Green, Yellow, Orange, White, Red) with height range and Roman numeral; three for a new one
☐	2.3	Tap Green, Yellow, Blue	Session total rises by one each tap; Your record today below the strip shows the three, newest first, with time, height and gauge

☐	2.4	Tap Undo last	Total goes down by one; the undone spring is named
☐	2.5	Gauge picker	OSG-01, OSG-02, ISG-01, SSG-02 named. Position Snubber → SSG-02 → amber "no calibration date is recorded"
☐	2.6	The red bar under the strip: Condemn this spring	A reason must be chosen (crack, corrosion, deformation, under height...); a photograph is asked for if photographing is on; recorded CONDEMNED with the reason and appears red in Your record today . Cancel and confirm both work
☐	2.7	Tick Photograph springs while sorting	The live camera opens right under the tick box, large, with a green <i>Photographing</i> badge. Nothing else to choose — keep tapping bands, each tap saves a photo. Hand in front → "person excluded"; never "spring 98%". <i>Teach the camera (advanced)</i> is folded away and not part of the day's work
☐	2.8	Bogie type LWLH25 or LCCF20	The strip disappears; a height is typed; the condemning height shown is from WMM §309C / G-112
☐	2.9	Hindi toggle here	Band names in Hindi, same colours, numbers unchanged
☐	2.10	Single Spring — one spring <i>on a wagon</i> , so it first asks which wagon and bogie; scroll to Free height	258.5 NLB outer used → Green, Table 28. 244 → CONDEMNED with reason. 263.1 → over-height, condemned. "abc" → refused, no crash
☐	2.11	A wagon → All wagons	The wagons in the shop with their stage. Type WR/BCNHL/40112 → it opens
☐	2.12	Wagon → Checklist	Category tabs; Pass • Fail • Condemned • Repaired • Replaced per item; a condemned item asks for a note

☐	2.13	Wagon → Photos	Four frames with pocket counts. Bogie 1·A: two counts agree. Bogie 2·B: "blind recount by someone else is needed" and no figures . Open the counter: the expected number is not shown
☐	2.14	Wagon → Parts — <i>Parts in, parts out</i> : what came off, what went back on, what was scrapped, by position	A balance line (<i>40 off · 40 back on ...</i>); in the list one outer spring Replaced with new — crack at the second coil ; the form offers Came off · Went back on · Replaced with new · Scrapped · Not being refitted , then Category, Position, Part, How many
☐	2.15	Wagon → Air-brake test tab (the Single Wagon Test)	The 14-row proforma; each row judged against the WMM limit as you type
☐	2.16	Wagon → Timeline	Stage history with dates and who moved it
☐	2.17	SER/BOXNHL/30914 → Checklist → Wheels → tap a wheel	One large box: the tread diameter chalked on the disc. Type 917 → <i>below issue limit</i> against 919 / 906, drawing and rule beneath. Flange figures are folded under <i>optional</i> — only if the tyre defect gauge was used
☐	2.18	Read a wagon number with the camera — point at the eleven stencilled digits (print 22057123457 large; it is a valid BOXNHL number). Not the SER/BOXNHL/... form, which is typed	It proposes the digits and the type they imply; you confirm or correct — never records unconfirmed. Letters and slashes are never read; a number whose check digit fails is not offered

☐	2.19	Manual	"brake block condemning limit" → Answer — in the document's own words , large, with <i>Source</i> : the manual and page beneath; more under <i>Also in</i> . "wheel diameter variation same axle bogie" → IRCA Part III by name. It quotes, never paraphrases — deliberate
☐	2.20	Wi-Fi off. Sort three springs. Wi-Fi on.	They sort offline (queued count shows); within a minute the queue empties and exactly three appear in the laptop's records, no duplicates
☐	2.21	Type the DRM dashboard or Audit URL by hand	Refused — an inspector does not have those screens
☐	2.22	Sign out	Sign-in page; the back button does not reopen a signed-in screen

3. Supervisor — on the laptop (projector)

0 / 17

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Sign in as `supervisor1`

☐	3.1	Wagons Pipeline	Thirteen wagons across seven stages; stage, days in shop and target date on each card
☐	3.2	Register a wagon	Number, type, railway; a duplicate number refused; a nonsense type refused
☐	3.3	WR/BCNHL/40112 → Photos → Pocket counts	The screen the inspector saw in 2.13, as a supervisor: three frames complete, and on Bogie 2 • Side B the figures the inspector was not shown — <i>6 of 7 outer counted</i> — because a supervisor may see a first count before the blind recount

☐	3.4	→ Release checks	The short count listed as an advisory, by name; release needs it acknowledged
☐	3.5	SER/BOXNHL/30914 → Release checks	Blockers: a condemned spring; axle 3 wheels at 917.5 mm below the 919 shop-issue diameter; one bearing rotation check pending. Authorise is disabled
☐	3.6	→ Checklist → Wheels	Eight chalk figures with limit and drawing beside each; axle 3 flagged
☐	3.7	→ Condition report	A printable one-page report
☐	3.8	SECR/BOXNHL/10492 (released) → Release Certificate	Number, date (weeks ago, not today), TAT, the eight categories, a QR. Scan the QR with the iPad's Camera app (same hotspot): it opens /verify.html on this server and shows VERIFIED . The QR carries the Mac's Wi-Fi address even when opened from localhost. Print gives a clean page
☐	3.9	→ Passport → Export the passport	A .jsonl file downloads — for the next workshop's software , not for reading. Under the button: <i>What the passport holds</i> — stage moves, spring readings, checklist verdicts, parts, tests, photographs, the sign-off, with dates and who sealed it
☐	3.10	Pick a new date in the <i>Due out</i> box on a wagon in progress	A box asks why ; <i>Move the date</i> stays disabled until a reason is typed; then a green line: <i>Due-out moved 19 → 22, with the reason, on the audit trail</i>
☐	3.11	<i>Stores, Passports, History, Analytics, Learning are not on the supervisor's menu</i>	By design: the pilot keeps the supervisor's menu to the shop floor (pilotScope.ts). They are tested under DRM (4.3, 4.4) and Admin (5.9–5.11)

☐	3.15	Ask the Manual	As 2.19
☐	3.17	Shadow Run	A week of the log; the verdict; "the snubber on NR/BOXN/60334 — the register would have passed it". Record a discrepancy and a summary; both appear
☐	3.18	Ask the Records	"which wagon type condemns the most snubbers this quarter" → answer with count → How this was computed shows SQL and rows. "is any gauge reading high" → OSG-02, 1.1 mm. A question it cannot answer → it says so
☐	3.19	Audit Chain → Verify chain again	The green card pulses once and gains <i>Re-derived just now — at HH:MM:SS, in N ms (check 2 this session)</i> ; the count of entries checked
☐	3.20	Single Spring → wagon, bogie, height → at the bottom next to Save, the amber Supervisor override — change the band	Asks the new band, a reason and the one-time code; the record shows both the original and the overridden band
☐	3.21	Hindi toggle on pipeline and release checks	Everything translated; wagon numbers and figures unchanged

4. DRM — on the laptop

0 / 9 [Clear this section](#)

Sign in as [drm1](#)

☐	4.1	DRM Dashboard — Shop Floor, Right Now	Pile finish, bogies buildable, what Stores will need — each with its basis, none "Not yet known"
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☐	4.2	Scroll the dashboard	Turnaround, where wagons wait, what keeps coming back, inspector quality, gauge exposure — every rate with its count; the band pie shows six colours and includes the bench
☐	4.3	Spring Analytics	Bench totals today/week; tap a day's bar → that day's figures read beneath; stock by band; the standard report with n on every figure; the Gauge drift panel naming OSG-02
☐	4.4	System Learning → Run Analysis	Each subsystem with agreement and count; the camera says it has been taught nothing; auto-commit off. After <i>Run Analysis</i> a green line: <i>Analysed at HH:MM — N observations, 0 new proposals (nothing worth changing yet)</i>
☐	4.5	Shadow Run	Visible, read-only — no record buttons
☐	4.6	Ask the Records	As 3.18
☐	4.7	Audit Chain → Verify	As 3.19
☐	4.8	Wagons Pipeline	No Register New Wagon button for the DRM, and the server refuses the DRM too. (As supervisor: a junk number like ADSFADS is refused — "not a wagon number"; a wagon registered by mistake → open it → <i>Registered in error?</i> → reason → one-time code → it leaves the pipeline; the record stays)
☐	4.9	Open SECR/BOXNHL/10492 → Release Certificate → scan the QR with the iPad (same hotspot)	The phone opens the verify page and shows VERIFIED — the signature checked against the shop's key in the phone's browser. The same page also takes a certificate <i>file</i> with no server at all — what another railway would do

5. Admin — on the laptop

0 / 11

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Sign in as **admin1**

☐	5.1	User Accounts	Create an account (one-time code asked); deactivate → cannot sign in; reactivate → can
☐	5.2	Roster import — three ways: one at a time (form at the top), paste from a spreadsheet, or open the office's CSV file	Paste or open three lines (name, ID, role) → preview → confirm → three printable slips with passwords
☐	5.3	<i>What each role holds</i> table	Read-only by design (the note above says why): who may do what, fixed in the system; the admin cannot release a wagon. To change a person's access, change their role on their row
☐	5.4	Checklist Rules	The blue box says what it is, what you can do, and when. Pick a wagon type; add a shop line with reason and source; withdraw it — each recorded under your name
☐	5.5	Gauge Register (bottom of User Accounts) — the bench instruments and their calibration papers	Four gauges; SSG-02 shows its certificate number and <i>NOT RECORDED</i> ; <i>Record calibration</i> → the dates from certificate 1251122-04-125 → amber goes. The drift box lists one line per finding : "the two gauges on this kind disagree by 1.5 mm — put both against the master"
☐	5.6	DRM Dashboard as admin — <i>Deployment readiness</i> and <i>Storage</i> are the first two panels	Every row a plain sentence: backup, cloud copy, manual (green, each document named), demo passwords (red on the demo record, saying why they still sign in), restarts, audit chain, storage

☐	5.7	Top right: Password (key icon, now labelled)	The panel opens in full over the page (it was cut off under the header). Old password stops working; new one works. Same for every role
☐	5.8	Top right: Authenticator (shield icon, now labelled)	The panel opens in full; a QR to scan; wrong code refused; right code enrolls; sign-in then needs the code
☐	5.9	Stores & Inventory	+ Add a part → code, name, category, bin → it appears in the list; restock 10 → stock rises by 10; restock 1.5 → refused
☐	5.10	Component Passports	Paste WRS-PASSPORT WHL-RWF-2023-8841 WHEELSET RWF_YELAHANKA → its history
☐	5.11	History & Logs	Pick a band or status → the list narrows at once; type part of a wagon number → narrows as you type; the count line says <i>N records matching ...</i> with <i>Clear filters</i> . Export Audit Trail (top bar) asks for the one-time code

6. Failure drills — once, on the machine you are rehearsing on

0 / 6 [Clear this section](#)

On the Mac each is one command in a terminal in the project folder (`cd ~/Desktop/WRS_Raipur`); on the demo laptop it is one action. **start** = `bash scripts/rehearsal.sh start` (laptop: double-click START.cmd) · **stop** = `... stop` (laptop: close the START.cmd window) · **status** = the two addresses · **log** · **backup** · **restore** · **tunnel**

☐	6.1	Phone: sign in as inspector1, open Sorting. Mac: bash scripts/rehearsal.sh stop. Phone: tap two bands.	The phone shows offline / waiting to send; the taps are counted. Then <code>bash scripts/rehearsal.sh start</code> (a minute or two). The phone reconnects on its own; the queue empties; Your record today shows the two taps once each
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☐	6.2	Mac: turn Wi-Fi off (not <i>stop</i> — the server keeps running, the network goes). Watch the phone	The phone shows Offline and keeps sorting; the Mac's own screen at localhost keeps working. Wi-Fi on → the phone reconnects within seconds. (If instead you <i>stop</i> the server, every screen says <i>Cannot reach the server — it may be restarting...</i> , not "Failed to fetch")
☐	6.3	<code>bash scripts/rehearsal.sh log</code>	Lines for every start; every request with its status; the stop from 6.1 written with its reason
☐	6.4	<code>bash scripts/rehearsal.sh backup</code>	A ...db.enc and its .hmac listed in backups-elsewhere/ with the time. As admin, the readiness row for the backup says <i>newest 0 hours ago</i>
☐	6.5	<code>bash scripts/rehearsal.sh restore</code>	The newest backup restored into a second folder and a table — wagons, inspections, sorting records, audit entries — live and restored counts equal
☐	6.6	<i>Plan B only.</i> <code>bash scripts/rehearsal.sh tunnel</code>	A public https://...trycloudflare.com address that opens on any phone with no certificate warning. Demonstration only — the data crosses Cloudflare — never for the shop's live record

What must not happen, anywhere

- No screen ever says "spring 98%" or shows a confidence number a person did not enter.
- No inspector ever sees a pocket count's figures before the blind recount.
- No record can be edited or deleted — only added to. Try; it refuses.
- No page reaches the internet. In the browser's network panel every request goes to localhost or the laptop's own address.
- Nothing asks for an email, a phone number, or a photograph of a person.

